microDX21

Yamaha DX21 Emulator

OWNER'S MANUAL

Raspberry Pi 3 / 4 / 5 · Bare-Metal · KY-040 Rotary Encoder

This manual describes how to operate microDX21 with the rotary encoder and the 128x32 OLED display.

Functions mirror the original Yamaha DX21 Owner's Manual; strings are extracted from the disassembled ROM V1.5 firmware.

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1. Operating the Rotary Encoder

Unlike the original, the microDX21 has only a single input device: a KY-040 rotary encoder with push-button. Every DX21 front-panel control (PLAY/EDIT/FUNCTION/PERFORMANCE/COMPARE/STORE/Cassette/Memory Protect, …) is folded onto that one knob. The binding follows the six modes of the original firmware, whose mode→display tables are extracted verbatim in `src/opm/dx21_ui_strings.h`.

1.1 Gesture Reference

Gesture	Effect
Rotate CW / CCW	Value +1 / -1 per detent
Single click	Next mode: PLAY → EDIT → PERFORM → FUNCTION → MEMORY → PLAY
Double click	COMPARE: audition the original / buffer voice (PLAY/EDIT/PERFORMANCE)
Triple click	In EDIT mode: cycle the operator OP1 → OP2 → OP3 → OP4
Hold 1 s (first tick)	Toggle between Browse and Edit
Hold 2 s (second tick)	Toggle MEMORY PROTECT

```
Display (128 x 32 OLED, pages 0..3):
+-------------------------------------+
| MODE TITLE                 16/36    |  Page 0
|   42          7-segment value       |  Page 1
| PARAMETER NAME                      |  Page 2
| STATUS / HINT                       |  Page 3
+-------------------------------------+
```

1.2 Browse vs. Edit Mode

When you enter a new mode the display starts automatically in **Browse mode** (status line shows `BROWSE`): rotation navigates the parameter list without touching the synth. The first hold (1 s) switches to **Edit** (status `EDIT`): rotation now changes the value of the currently selected parameter.

The split is necessary because a single encoder must handle both list navigation and value editing. The mode list is one-dimensional: **EDIT** and **FUNCTION** prefer value-editing; **PLAY**, **PERFORMANCE** and **MEMORY** work in browse (list selection).

> *Note: in PLAY and PERFORMANCE the encoder is always live — rotation changes the current program instantly. In MEMORY rotation moves through the action / picker list. Hold reaches only the Memory-Protect toggle here.*

1.3 Status-Line Messages

The status line (page 3) shows transient feedback. Important messages:

Message	Meaning
BROWSE / EDIT	Mode status after a 1 s hold toggle
MEMORY PROTECTED/UNPROTEC.	Hold-2 s toggle of Memory Protect
COMPARE:ORIGINAL	Double-click engaged COMPARE
NO EDITS YET	Double-click refused — the voice has not been edited yet
EDIT OP1..OP4	Triple-click switched the target operator
FROM OPx>OPy	EG Copy: rotation is picking the source operator

EG OPx>OPy OK	EG Copy executed
SRC=DST OPx	EG Copy refused: source and destination are the same
BULK TRANSMITTED / BULK FAILED	SysEx bulk dump sent / failed
NAME EDIT/STORED/REJECTED	Voice-name editor active / committed / refused
MIDI BUFFER FULL/MIDI CSUM ERROR	MIDI receive-buffer error

2. PLAY Mode — Performing

PLAY corresponds to the SINGLE / DUAL / SPLIT block of the original manual. The display shows the number and name of the current voice and the active play-mode label (`SINGLE`, `DUAL` or `SPLIT`).

```
PLAY view:
+------------------------------+
|  <042>                       |  Voice number
|  STRINGS                     |  Voice name (inverted under COMPARE)
|  A1-A8                       |  Bank (1 of 4, derived from voice number)
|  PLAY  SINGLE                |  Mode label
+------------------------------+
```

2.1 SINGLE / DUAL / SPLIT

The play mode is changed in FUNCTION mode (entries **Poly Mode** / **Mono Mode**, or the voice-mode parameter directly). In **DUAL** the two halves can carry independent voices; **SPLIT** divides the keyboard at the split point into a left and right side. In SPLIT the Mono/Poly assignment is taken from each side's patch (VCED byte 63), so layouts are 4+4, 7+1, 1+7 or 1+1.

2.2 Voice Selection 1..128

Rotation picks the next / previous voice. Voices 1..128 are organised as 16 groups of 8 (Group 1..16, 8 voices each). The current bank (A1-A8, A9-A16, B1-A8, B9-B16) is shown on page 2. The RAM range (voices 1..32) holds your own patches; voices 33..128 are the DX21 factory ROM sounds.

> *Tip: once the MIDI Sy Info flag is set (FUNCTION #5), every voice change additionally sends a 1-voice VCED dump (F0 43 2n 03 .. F7) out of MIDI OUT.*

3. EDIT Mode — Voice Programming

EDIT exposes all 36 DX21 voice parameters. The order matches the list in `dat_F610` of the original firmware. Page 1 shows the current value as a 3-digit 7-segment number (0..255, interpreted as 0..127 / 0..99 / 0..7 depending on the parameter).

3.1 EDIT Parameters (Table)

#	Name	Range	Effect
1	ALG	0..7	Algorithm
2	ALGORITHM SELECT	0..7	Algorithm (long form)
3	FEEDBACK	0..7	Feedback level
4	LFO WAVE:	0..3	LFO waveform (TRIANGLE / SAW UP / SQUARE / S/HOLD)
5	LFO SPEED	0..99	LFO speed
6	LFO DELAY	0..99	LFO onset delay
7	LFO PMD	0..99	LFO → pitch mod depth
8	LFO AMD	0..99	LFO → amp mod depth
9	P MOD SENS.	0..7	Pitch-mod sensitivity
10	A MOD SENS.	0..3	Amp-mod sensitivity
11	E BIAS SENS.	0..7	EG-bias sensitivity (OP)
12	KEY VELOCITY	0..7	Velocity sensitivity (OP)
13	FREQUENCY =	0..63	Coarse ratio (OP)
14	DETUNE =	-7..+7	Detune in cents (OP)
15	EG AR	0..31	Attack rate (OP)
16	EG D1R	0..31	Decay-1 rate (OP)
17	EG D1L	0..15	Decay-1 level (OP)
18	EG D2R	0..31	Decay-2 rate (OP)
19	EG RR	0..15	Release rate (OP)
20	OUTPUT LEVEL	0..99	Operator level (OP)
21	RATE SCALE	0..3	Rate scaling (OP)
22	LEVEL SCALE	0..99	Level scaling (OP)
23	PEG RATE 1	0..99	Pitch-EG rate 1
24	PEG LEVEL 1	0..99	Pitch-EG level 1
25	PEG RATE 2	0..99	Pitch-EG rate 2
26	PEG LEVEL 2	0..99	Pitch-EG level 2
27	PEG RATE 3	0..99	Pitch-EG rate 3
28	PEG LEVEL 3	0..99	Pitch-EG level 3
29	SAW UP	—	Waveform alias
30	SQUARE	—	Waveform alias
31	TRIANGL	—	Waveform alias
32	S/HOLD	—	Waveform alias
33	EG Copy	Utility	EG Copy: rotation picks the source OP
34	from OP	Utility	EG Copy: current source-OP selection
35	LFO SYNC :	ON/OFF	LFO key sync
36	Memory Protected	ON/OFF	Memory Protect (see Ch. 8)

3.2 Operator Selection (OP1..OP4)

Parameters 11..22 and 23..27 (per-operator: EG, OUT, FREQ, DET, RS, LS, EBS, KVS) act on the currently selected operator. In EDIT mode the selection is reached with a **triple click**: OP1 → OP2 → OP3 → OP4 → OP1. The current selection is shown in the title as `OPn` and acknowledged on the status line whenever it changes.

3.3 EG Copy (Utility)

EG Copy duplicates the five EG parameters (AR, D1R, D1L, D2R, RR) from a source operator into the destination operator chosen via triple-click. Procedure:

- Triple-click in EDIT mode → pick the destination OP (status: `EDIT OP1..4`).
- Navigate to entry **EG Copy** or **from OP**.
- Rotate → set the source OP (status: `FROM OPx>OPy`).
- Single click → run the copy. On success the status line shows `EG OPx>OPy OK`; otherwise `SRC=DST OPx` or `EG COPY FAILED`.

> *Source and destination must differ. With COMPARE active the snapshot is renewed automatically; the Memory-Protect gate applies here as well — if it is ON, the copy is refused.*

4. PERFORMANCE Mode — 32 Performance Memories

PERFORMANCE addresses the 32 performance memories (slot 1..32). Each performance stores: play mode (SINGLE/DUAL/SPLIT), voice A, voice B, split point, balance, pitch-bend range and mode, portamento mode and rate, mod-wheel and breath-controller sensitivities, chorus, transpose, key shift and a 10-character name.

```
PERFORMANCE view:
+------------------------------+
| PERFORM SI  3/32             |  Mode tag, slot
|  CHOIR                       |  Performance name (or BUFF under COMPARE)
| A:005  B:008                 |  Programmed voices
| EDIT=APPLY                   |  Hint
+------------------------------+
```

4.1 Selecting a Performance

Rotation in Browse mode steps through slot 1..32. The status line shows `PERF APPLIED` if a valid slot was loaded, otherwise `EMPTY PERF`. In COMPARE mode the slot name is replaced by `BUFF` (see Ch. 7).

4.2 Saving a Performance

Saving happens through the **MEMORY** mode (action **PERF**, see Ch. 6). The selected performance is held in the edit buffer (*ApplyPerformance*) until you explicitly store it.

5. FUNCTION Mode — Configuration

FUNCTION corresponds to the FUNCTION block of the original manual. All 46 entries from `dat_FBDA` are preserved in the same order, so the assignment to keys, actions and values stays identical.

5.1 Tuning (Master Tune, Dual Detune)

#	Name	Values	Effect
1	Master Tune =	-64..+63	Tuning in cents; A3 = 440 Hz at 0
2	Dual Detune	0..99	Detune of the B side in DUAL mode

5.2 MIDI Functions

#	Name	Values	Effect
3	Midi Switch :	ON / OFF	MIDI receive master switch
4	Midi Ch Info:	ON / OFF	Channel information (CC/PC)
5	Midi Sy Info:	ON / OFF	SysEx information (real-time param change, bulk)
6	Midi Recv Ch =	1..16	Receive channel
7	Midi Omni :	ON / OFF	OMNI mode
40	Midi Trns Ch =	0..15	Transmit channel (0 = off)
41	Midi Transmit ?	Action	Trigger a bulk dump over MIDI OUT

5.3 Performance and Foot-Switch Functions

#	Name	Values	Effect
8	Recall Edit ?	Action	Edit Recall: bring back the last edited voice
9	Are You Sure ?	Action	Confirmation dialog
10	Init. Voice ?	Action	Reset all voice parameters to their initialised values
24	Poly Mode	—	Switch play mode SINGLE / DUAL / SPLIT
25	Mono Mode	ON/OFF	Mono operation (SINGLE only)
26	P Bend Range	0..12	Pitch-bend range in semitones
27	Fingered Porta	ON/OFF	Fingered portamento
28	Full Time Porta	ON/OFF	Full-time portamento
29	Porta Time	0..99	Portamento rate
30	Foot Volume	0..99	CC#7 volume range
31	Foot Sustain:	ON/OFF	CC#64 sustain-pedal enable
32	Foot Porta :	ON/OFF	CC#65 portamento-foot-switch enable
33	MW Pitch	0..99	Mod wheel → pitch sensitivity
34	MW Amplitude	0..99	Mod wheel → amp sensitivity
35	BC Pitch	0..99	Breath controller → pitch sensitivity
36	BC Amplitude	0..99	Breath controller → amp sensitivity
37	BC Pitch Bias	0..99	Breath controller → pitch bias
38	BC EG Bias	0..99	Breath controller → EG bias
39	Middle C =	-24..+24	Key offset
42	Chorus :	ON/OFF	3-line ensemble / chorus
43	Bend Mode =	0..3	Pitch-bend mode (All / Low / High / K-on)

44	Key Shift =	-24..+24	Transpose in semitones
45	Name :	Editor	Voice-name editor (see 5.4)

5.4 Voice-Name Editor

Entry 45 (`Name :`) opens a 10-character editor. Rotation changes the character at the cursor position (character set: `A–Z, a–z, 0–9, space, '-', '.', '/'`); a short click advances the cursor. After the 10th position the name is committed. The status line acknowledges with `NAME STORED`, or — if Memory Protect is ON — `NAME REJECTED`.

6. MEMORY Mode — SD Card / Performance Store

MEMORY is a five-stage state machine that maps the original Cassette / Bank functions onto the SD card. There are seven actions; each is operated by an action picker (stage 0), a YES/NO confirmation (stage 1), up to two value pickers (stage 2/4) and a result display (stage 3).

```
MEMORY view:
+-------------------------------+
| MEMORY  ACT 1                 |  Stage 0: pick action
|  SAVE                         |  Action name (large)
| Save to SD    ?               |  Hint text
| (Status / empty)              |
+-------------------------------+
```

6.1 The Seven Actions

#	Action	Picker 1	Picker 2	Effect
1	SAVE	Group 1..16	—	32 RAM voices → SD:/MICRODX21/BANK_NN/
2	LOAD	Group 1..16	—	SD bank → 32 RAM voices
3	VRFY	Group 1..16	—	Verify SD bank vs. RAM
4	STOR	Slot 1..32	—	Store the edit voice into a RAM slot
5	ROMG	Group 1..16	Bank 1..4	ROM group (8 voices) → RAM bank
6	ROMV	Voice 1..128	Slot 1..32	One ROM voice → edit buffer → RAM
7	PERF	Slot 1..32	—	Save the current performance

6.2 Worked Examples

Example A: save the current 32 RAM voices to SD

- Rotate → pick **SAVE** (no. 1).
- Click → YES/NO. Rotate to YES.
- Click → Group picker 1..16; rotate to choose (e.g. 1).
- Click → execution. The display briefly shows `OK` or `SAVE FAILED`, then returns to stage 0.

Example B: load a ROM group into a RAM bank

- Rotate → pick **ROMG** (no. 5).
- Click → YES.
- Click → Group picker (ROM group 1..16). Rotate, click.
- Click → Slot picker (RAM bank 1..4 = A1-A8 / A9-A16 / B1-A8 / B9-B16). Rotate, click.
- Click → execution. Memory Protect must be OFF.

> *Before any write action (STORE, ROMG, ROMV, PERF) Memory Protect must be OFF. When the write-protect is active the call is aborted and the status line shows* `MEMORY PROTECTED` *(Ch. 8).*

7. COMPARE — Original vs. Edit

While editing, the original (or the voice in the edit buffer) can be auditioned directly without losing the in-progress edits. **Double-click** in PLAY, EDIT or PERFORMANCE toggles COMPARE.

In COMPARE state:

- In PLAY the voice name is drawn inverted;
- In EDIT and PERFORMANCE the voice / performance slot is replaced by `BUFF`;
- The title row additionally shows the COMPARE marker;
- The engine switches back to the snapshot taken before the edit; `setCompare(true)` writes a snapshot as soon as the first edit change arrives.

> *Refused with status* `NO EDITS YET` *if no edit change has happened yet. The original manual also specifies this rule.*

8. MEMORY PROTECT

Memory Protect prevents accidental overwriting of the 32 RAM voices and the 32 performance memories. A 2×2 block in the top-right corner of the title row is shown whenever Protect is active.

To toggle:

- Hold 2 s (second hold tick) — mode-independent.
- Status line: `MEMORY PROTECTED` / `MEMORY UNPROTECTED`.
- As an alternative, EDIT entry 36 also toggles Protect from the list display.

Write paths that respect Memory Protect: **STORE Voice** (MEMORY #4), **ROMG / ROMV** (#5/#6), **PERF** (#7), **set voice name** (FUNCTION #45), and all 32-voice bulk dumps received via SysEx.

9. MIDI Data Format (Summary)

microDX21 is hardware-DX21-compatible at the SysEx level. The following formats are supported:

Type	Format	Hex
Real-time parameter change	F0 43 1n 12 pp vv F7	0x10 + n
1-voice VCED dump	F0 43 2n 03 ... F7	f=3
32-voice VMEM bulk	F0 43 2n 04 ... F7	f=4
Dump request	F0 43 2n ff F7	f=3/4

Real-time parameter change sends VCED parameter numbers 0..92 (operators + global voice parameters) plus parameter 93 (operator-enable bit field). The panel-edit routine uses the same numbering; incoming MIDI edits never echo back (*applySysexParam* sets *transmit=false*).

When selecting a voice in PLAY SINGLE the matching 1-voice VCED dump is additionally sent out of MIDI OUT (gated on `FUNCTION #5 Midi Sy Info`).

The **32-voice bulk** sequence sends 4096 data bytes (32 × 128, 73 bytes of packed VMEM + 55 zero padding per voice) with a two's-complement checksum. On receive the legacy format f=0x09 (76-byte VCED, simple sum) is also accepted for maximum interoperability with older editors.

On receive the **edit-slot** voice (`m_sysexEditVoice`) is the destination. It can be set by MIDI Program Change (voice 1..32 = RAM slot; voices from the ROM are read-only until they are copied into the edit slot via the MEMORY action ROMV).

10. Factory Defaults and Bill of Materials

10.1 Factory Defaults

Parameter	Value
Play mode	SINGLE
Poly / Mono	POLY
Master Tune	0
MIDI Switch	ON
Receive channel	1
Transmit channel	1
Sys Info	ON
CH Info	ON
Memory Protect	ON (turn OFF for a load, then back ON)
Chorus / Ensemble	ON
Pitch Bend Range	2
Foot Volume / Sustain / Porta	OFF

10.2 Bill of Materials and Prerequisites

- Raspberry Pi 3, 4 or 5 (BCM2837/2711/2712) — no Linux, Circle bare-metal build.
- I2S DAC PCM5102A (or on-board PWM) — configured in `config/microdx21.ini`.
- 128 × 32 OLED: SSD1305 / SH1106, SPI or I2C, DC pin 24, reset 25, SPI0 8 MHz (default).
- KY-040 rotary encoder — CLK / DT / SW on GPIO 10 / 9 / 11 (default, editable in `microdx21.ini`).
- USB-MIDI host (DIN adapter) or — on Pi 4 / 5 — the RP2350 Comms Pi over UART GPIO 14 / 15.
- microSD card with boot firmware (`kernel8.img` for Pi 3, `kernel8-rpi4.img` for Pi 4, `kernel_2712.img` for Pi 5) and `config/microdx21.ini`.

10.3 Gesture Cheat-Sheet

Gesture	Effect (across all modes)
Rotate	Value ±1 / list navigation / picker change
Single click	Next mode (PLAY → EDIT → PERFORM → FUNCTION → MEMORY → PLAY)
Double click	COMPARE on / off (PLAY / EDIT / PERFORM)
Triple click	OP1→OP2→OP3→OP4 (EDIT only)
Hold 1 s	Browse ↔ Edit (only meaningful in EDIT / FUNCTION)
Hold 2 s	Toggle MEMORY PROTECT

Appendix A: DX21 Button → Encoder Gesture Mapping

The original DX21 has more than 30 buttons (PLAY SINGLE / PLAY DUAL / PLAY SPLIT / EDIT / COMPARE / FUNCTION / 16 voice selectors A1..A8 + B1..B8 / DATA ENTRY / -1 / +1 / STORE / PB MODE SET / KEY SHIFT / CHARACTER). The table below shows how every one of them maps onto the single encoder.

DX21 control	Encoder gesture
PLAY SINGLE / DUAL / SPLIT	FUNCTION mode: entry 24 / 25 (Poly / Mono mode & voice-mode parameter)
EDIT	Single click cycles into EDIT mode
EDIT / COMPARE (second press)	Double click — COMPARE
FUNCTION	Single click cycles into FUNCTION mode
VOICE 1..32 (A1..A8 / B1..B8)	PLAY mode: rotation selects 1..128 (all 16 groups × 8)
VOICE 1..16 (PERFORMANCE)	PERFORMANCE mode: rotation selects slot 1..32
DATA ENTRY dial	Rotation in EDIT / FUNCTION (Edit mode)
DATA ENTRY +1 / -1	Rotation (one detent per direction)
STORE	MEMORY mode: action STOR, then slot picker, click
MEMORY PROTECT	Hold 2 s — status line acknowledges
PB MODE / KEY SHIFT	FUNCTION entry 43 (Bend Mode) and 44 (Key Shift)
CHARACTER (voice name)	FUNCTION #45 opens the name editor; rotation changes the character, click advances the cursor
Cassette SAVE / LOAD / VERIFY	MEMORY mode, action SAVE / LOAD / VRFY

Appendix B: Display Layout Reference

PLAY

```
Page 0: '  <NNN>'   voice number
Page 1: voice name (inverted under COMPARE)
Page 2: bank (A1-A8 / A9-A16 / B1-A8 / B9-B16)
Page 3: 'PLAY  MODE_LABEL' or status
Top-right pixels: COMPARE marker (4x2) + MEMORY PROTECT (2x2)
```

EDIT

```
Page 0: 'EDIT  OPn N/36' or 'EDIT  N/36'
Page 1: 3-digit 7-segment value (0..255)
Page 2: EDIT_PARAM_NAMES[N]
Page 3: status / hint / override
```

PERFORMANCE

```
Page 0: 'PERFORM  SI  N/32'  (mode tag: SI / DU / SP)
Page 1: performance name (or BUFF under COMPARE)
Page 2: 'A:NNN  B:NNN'  (or '-- EMPTY --')
Page 3: status / hint
```

FUNCTION

```
Page 0: 'FUNCTION  N/46'
Page 1: function name (7-seg, 16 px high)
Page 2: '=NN' / 'ON' / 'OFF' / name editor / '=n/a'
Page 3: status / hint
```

MEMORY

```
Page 0: 'MEMORY  ACT N' / 'YES/NO' / 'PICK NNN' / 'DEST NNN' / 'DONE'
Page 1: SAVE / LOAD / VRFY / STOR / ROMG / ROMV / PERF / YES/NO / digits / OK / FAILED
Page 2: hint text (per action)
Page 3: status / hint
```